

# Brock Martin Lab 2

## Memorandum

The Minnesota Twins has reached out to us to analyze pitching statistics to see if there are any predictive relationships between the type of pitch thrown based on the count and whether the batter gets a successful hit based on the type of pitch thrown and its speed. In short, there does not seem to be any significant relationships in determining whether a batter successfully hits the pitch based on pitch type and release speed. As well as the prediction in the pitch type thrown based on the count.

Starting with the no significant relationship between the probability of batters hitting the ball based on pitch type and release speed. From *Logistic Code 02*, we see that release speed is not a significant predictor in determining this relationship. This can be seen in *Logistic Figure 01* as different pitches have their own optimal release speed. Looking at *Logistic Code 03 & 04* shows us the model summary and the confidence interval of probabilities that batters hit the ball based on the pitch type thrown. We see that there are no significant predictors from *Logistic Code 03*. From the assumption that there are no significant predictors and looking at the confidence intervals from *Logistic Code 04*, it is safe to say that batters have a 50/50 chance of successfully hitting any pitch thrown.

As for the pitch type thrown based on the count. There does not seem to be a big difference between the pitches thrown based on the count. From *MM Code 03*, we see that change ups was thrown the least amount of times compared to Verlander's other pitches. This is what causes some weird outputs from *MM Code 02* where some log-odds values are in the hundreds. As log-odds approach large values, the probability of that event occurring approaches 100%. This can be seen within the confidence intervals from *MM Code 02*. Not to mention that for most counts each of the other three pitches are preferred more than the change up. Also based on the confidence intervals from *MM Code 02*, we also see that there are no one preferred pitch type during any count.

To recap, there are no significant relationship between the pitch type thrown and the count as well as a batter getting a successful hit based on pitch type thrown and its release speed.

## Appendix

### Log-Odds to Probability Converter

```
probs <- function(log_odds = 0){  
  return(  
    exp(log_odds)/(1+exp(log_odds))  
  )  
}
```

### Data Cleaning

```
obj <- matrix(  
  str_split(  
    VerlanderData$count,  
    "_",  
    simplify=TRUE  
  ),  
  byrow=FALSE, nrow=3839, ncol=2,  
  dimnames=list(c(1:3839), c("Balls", "Strikes"))  
)  
VerlanderData <- VerlanderData %>%  
  mutate(  
    Balls = as.factor(obj[, "Balls"]),  
    Strikes = as.factor(obj[, "Strikes"]),  
    isSL = case_when(  
      pitch_type == "SL" ~ 1,  
      pitch_type != "SF" ~ 0,)  
    isFF = case_when(  
      pitch_type == "FF" ~ 1,  
      pitch_type != "FF" ~ 0,)  
    isCU = case_when(  
      pitch_type == "CU" ~ 1,  
      pitch_type != "CU" ~ 0,)  
    isCH = case_when(  
      pitch_type == "CH" ~ 1,  
      pitch_type != "CH" ~ 0,)  
    Hit = case_when(  
      description == "hit_into_play" ~ 1,  
      description != "hit_into_play" ~ 0)  
  )
```

### Multinomial Model

```
test1 <- multinom(pitch_type~Balls*Strikes, data=VerlanderData)
```

### MM Code 01

```
## # weights: 52 (36 variable)
## initial value 5321.984052
## iter 10 value 4360.455351
## iter 20 value 4161.594132
## iter 30 value 4128.126095
## iter 40 value 4126.930858
## iter 50 value 4126.880007
## final value 4126.873321
## converged
```

```
test2 <- multinom(pitch_type~Balls+Strikes, data=VerlanderData)
```

```
## # weights: 28 (18 variable)
## initial value 5321.984052
## iter 10 value 4372.201474
## iter 20 value 4162.485254
## iter 30 value 4162.059703
## final value 4162.047213
## converged
```

```
test3 <- multinom(pitch_type~count, data=VerlanderData)
```

```
## # weights: 52 (36 variable)
## initial value 5321.984052
## iter 10 value 4363.662081
## iter 20 value 4172.433801
## iter 30 value 4127.999324
## iter 40 value 4126.927626
## iter 50 value 4126.869945
## final value 4126.869481
## converged
```

```
anova(test1, test2, test="Chisq")
```

```
## Likelihood ratio tests of Multinomial Models
##
## Response: pitch_type
##
```

|      | Model           | Resid. df | Resid. Dev | Test   | Df | LR stat. | Pr(Chi)      |
|------|-----------------|-----------|------------|--------|----|----------|--------------|
| ## 1 | Balls + Strikes | 11499     | 8324.094   |        |    |          |              |
| ## 2 | Balls * Strikes | 11481     | 8253.747   | 1 vs 2 | 18 | 70.34779 | 3.946282e-08 |

Since there is a significance in the interaction between Balls and Strikes. We must consider this connection. However, it would be simpler to use the whole count instead of separating the count into balls and strikes. As by completing every interaction term would eventually be similar to the whole count model.

```
summary(test3)
```

MM Code 02

```
## Warning in sqrt(diag(vc)): NaNs produced

## Call:
## multinom(formula = pitch_type ~ count, data = VerlanderData)
##
## Coefficients:
## (Intercept) count0-1 count0-2 count1-0 count1-1 count1-2 count2-0
## CU 1.603362 0.1439364 -0.5574095 -1.00864976 0.3061604 -0.3563451 35.97609
## FF 2.887009 -0.5844280 -1.5201383 -0.34639430 -0.3156847 -1.1360797 39.11120
## SL 1.546923 0.3215197 0.2575650 0.08721028 0.6315917 0.5274916 39.02818
## count2-1 count2-2 count3-0 count3-1 count3-2
## CU -1.8040478 -0.2170798 -4.782350 -18.38965 34.39509
## FF -0.6501787 -0.7198853 10.568584 38.18891 35.06749
## SL -0.1381479 0.6576570 -3.356967 37.00328 35.89000
##
## Std. Errors:
## (Intercept) count0-1 count0-2 count1-0 count1-1 count1-2 count2-0
## CU 0.1907900 0.2957272 0.2972804 0.3652169 0.3634261 0.2964703 0.6744753
## FF 0.1788643 0.2825212 0.2832975 0.3153070 0.3488464 0.2809562 0.3523598
## SL 0.1917156 0.2948632 0.2855612 0.3338412 0.3598900 0.2859712 0.3720155
## count2-1 count2-2 count3-0 count3-1 count3-2
## CU 0.4882853 0.3460240 5.814814 NaN 0.1958724
## FF 0.3641569 0.3260235 166.250527 0.2636929 0.1282737
## SL 0.3871510 0.3328260 25.470472 0.2636929 0.1425634
##
## Residual Deviance: 8253.739
## AIC: 8325.739
```

```
BigCov <- vcov(test3)
PureSE <- sqrt(diag(BigCov))
```

```
## Warning in sqrt(diag(BigCov)): NaNs produced
```

```
round(
  probs(
    t(
      rbind(
        coef(test3)[1,]-1.96*PureSE[1:12],
        coef(test3)[1,]+1.96*PureSE[1:12],
        coef(test3)[2,]-1.96*PureSE[13:24],
        coef(test3)[2,]+1.96*PureSE[13:24],
        coef(test3)[3,]-1.96*PureSE[25:36],
        coef(test3)[3,]+1.96*PureSE[25:36]
      )
    )
  ),3)
```

```
##           [,1] [,2] [,3] [,4] [,5] [,6]
## (Intercept) 0.774 0.878 0.927 0.962 0.763 0.872
## count0-1    0.393 0.673 0.243 0.492 0.436 0.711
## count0-2    0.242 0.506 0.112 0.276 0.425 0.694
```

```
## count1-0    0.151 0.427 0.276 0.567 0.362 0.677
## count1-1    0.400 0.735 0.269 0.591 0.482 0.792
## count1-2    0.281 0.556 0.156 0.358 0.492 0.748
## count2-0    1.000 1.000 1.000 1.000 1.000 1.000
## count2-1    0.059 0.300 0.204 0.516 0.290 0.650
## count2-2    0.290 0.613 0.204 0.480 0.501 0.788
## count3-0    0.000 0.999 0.000 1.000 0.000 1.000
## count3-1     NaN   NaN 1.000 1.000 1.000 1.000
## count3-2    1.000 1.000 1.000 1.000 1.000 1.000
```

```
VerlanderData %>%
  group_by(pitch_type) %>%
  summarize(num=n())
```

### MM Code 03

```
## # A tibble: 4 x 2
##   pitch_type  num
##   <chr>      <int>
## 1 CH         161
## 2 CU         651
## 3 FF        1914
## 4 SL        1113
```

### Predicting Hits based on pitch type and mph

```
VerlanderDataHit <- VerlanderData %>%
  filter(description == c(
    "hit_into_play", "swinging_strike", "swinging_strike_blocked"))
```

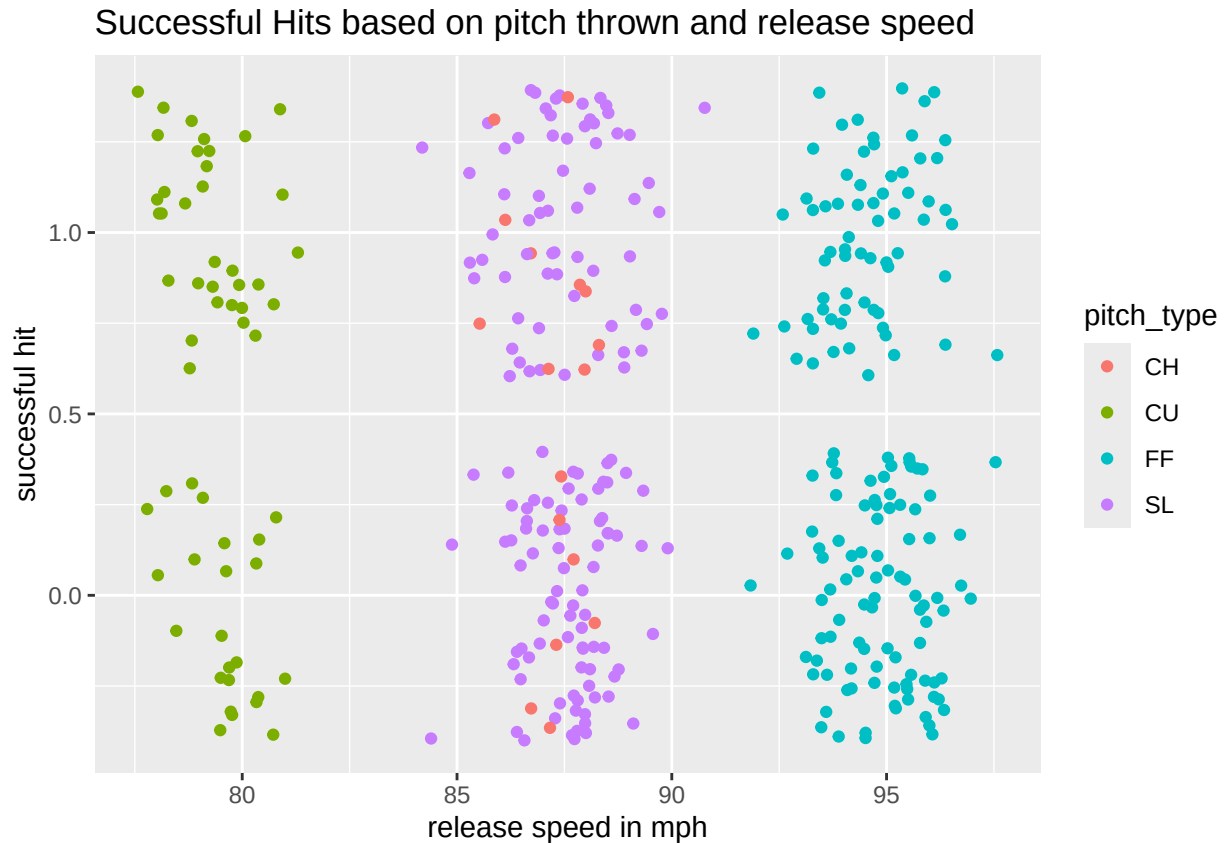
### Logistic Code 01

```
## Warning: There was 1 warning in 'filter()'.
## i In argument: 'description == c("hit_into_play", "swinging_strike",
##   "swinging_strike_blocked")'.
## Caused by warning in 'description == c("hit_into_play", "swinging_strike", "swinging_strike_blocked")'.
## ! longer object length is not a multiple of shorter object length
```

```
VerlanderDataHit %>%
  ggplot(aes(x=release_speed, y=Hit, col=pitch_type))+
  geom_jitter()+
  labs(
    title="Successful Hits based on pitch thrown and release speed",
    x="release speed in mph",
    y="successful hit",
    key="pitch type"
  )
```

## Logistic Figure 01

```
## Ignoring unknown labels:  
## * key : "pitch type"
```



```
HitModel <- glm(Hit~pitch_type+release_speed, family=binomial(), data=VerlanderDataHit)  
HitModel2 <- glm(Hit~pitch_type, family=binomial(), data=VerlanderDataHit)  
HitModel3 <- glm(Hit~release_speed, family=binomial(), data=VerlanderDataHit)  
anova(HitModel, HitModel2, test="Chisq")
```

## Logistic Code 02

```
## Analysis of Deviance Table  
##  
## Model 1: Hit ~ pitch_type + release_speed  
## Model 2: Hit ~ pitch_type  
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)  
## 1       377      517.05  
## 2       378      521.17 -1  -4.1155  0.04249 *  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(HitModel, HitModel3, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: Hit ~ pitch_type + release_speed
## Model 2: Hit ~ release_speed
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         377      517.05
## 2         380      521.47 -3   -4.4148    0.22
```

Since the reduced model is significant in reducing the deviance of the logistic model, we can say that release speed is not important in predicting whether or not a batter hits the ball thrown.

```
summary(HitModel2)
```

### Logistic Code 03

```
##
## Call:
## glm(formula = Hit ~ pitch_type, family = binomial(), data = VerlanderDataHit)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.35667    0.49281   0.724   0.469
## pitch_typeCU -0.03822    0.56109  -0.068   0.946
## pitch_typeFF -0.68458    0.51820  -1.321   0.186
## pitch_typeSL -0.60113    0.51989  -1.156   0.248
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 526.88  on 381  degrees of freedom
## Residual deviance: 521.17  on 378  degrees of freedom
## AIC: 529.17
##
## Number of Fisher Scoring iterations: 4
```

$$\log(\hat{\pi}_{hit}) = 0.35667 - 0.03822x_{\text{Change Up}} - 0.68458x_{\text{Fast Ball}} - 0.60113x_{\text{Slider}}$$

```
probs(confint(HitModel2, level=0.95))
```

### Logistic Code 04

```
## Waiting for profiling to be done...
```

```
##                2.5 %    97.5 %
## (Intercept)  0.3542854 0.7973319
## pitch_typeCU 0.2364607 0.7417244
## pitch_typeFF 0.1489678 0.5797979
## pitch_typeSL 0.1594604 0.6007906
```

$$H_O : Pr(Hit_{Pitch}) = 0.5$$

$$H_A : Pr(Hit_{Pitch}) \neq 0.5$$

Since all probabilities between every pitch type includes the 50% success rate, we fail to conclude that there is a significant predictor in determining whether a batter hits the ball based on pitch type.